CYBORG SUPERMARKET ENTERPRISE DOCUMENTATION

Complete Workflows and Sequence Diagram

Detailed Sequence Flows, Transitions, and Event Protocols

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Executive Publication Record

Organization:	Cyborg Omnichannel Engineering Organization
Ecosystem:	Windows POS • Android Mobile • Web Storefront • Admin Web • Backend
Primary Authority:	cyborg_system_architecture.pdf (16-Page Blueprint)
Source Dataset:	super_market(1).xlsx (1,000 Verified Egyptian Retail Records)
Security Level:	Enterprise Retail Distribution / Stakeholder Review
System Stack:	NestJS • TypeScript • PostgreSQL 16 • Redis 7 • SQLite • Flutter
Target Currency:	Egyptian Pounds (EGP) • Cairo Timezone (UTC+2 / UTC+3)

1. Offline POS Checkout & Cloud Ingestion Flow

The following sequence diagram outlines the deterministic synchronization cycle between the offline POS terminal and the central cloud database:

Step #	Component	Operation	Delivery Guarantee
01	Barcode Scanner	Cashier scans physical barcode	Hardware wedge interrupt
02	POS Local SQLite	Lookup cached product row and	Local memory (< 5ms)
03	Cash Drawer	Cashier confirms cash payment;	ESC/POS RJ12 interface
04	Local Outbox	Transaction inserted into sync	ACID local durability
05	Sync Engine	Background thread detects netw	At-least-once delivery
06	Backend Gateway	Idempotency check against even	Exactly-once processing

2. Order Lifecycle State Machine

All customer and POS orders obey a deterministic state transition hierarchy enforced on the server:

- PENDING -> CONFIRMED: Online order payment authorized (Card, Mobile Wallet, Fawry).
- CONFIRMED -> PREPARING: Warehouse / branch staff begin item assembly.
- PREPARING -> READY: Order packed and staged in collection area.
- READY -> DELIVERED: Order received by customer or handed to in-store shopper.
- DELIVERED -> REFUNDED: Authorized manager processes customer return with inventory restoration.

3. Egyptian Arabic Voice Shopping Pipeline

Voice queries in Egyptian dialect follow a 5-stage transformation pipeline:

- Speech Transcription: Extracts colloquial spoken Arabic ('???? ??? ???? ?????').
- Dialect Lexicon: Maps Egyptian grocery synonyms (??? -> ????, ???? -> ???).
- Database Matching: Scores and retrieves real catalog products with current stock status.